

# **Hybrid ConvNext and LLM Framework with Grad-CAM for Papaya Leaf Disease Detection and Treatment Recommendation**

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## **Abstract:**

Papaya is an economically important tropical fruit crop valued for its nutritional and medicinal benefits; however, its productivity is severely affected by leaf diseases such as leaf curl, powdery mildew, and leaf spot. In this study, an intelligent system is proposed for automated papaya leaf disease detection and management using advanced deep learning techniques. The proposed framework integrates Deep Learning, Explainable Artificial Intelligence (XAI), and a Large Language Model (LLM) Llama 3B to provide Treatment Recommendation . A ConvNeXt-based convolutional neural network is employed for disease classification from leaf images, along with severity assessment at three levels: initial, moderate, and severe. The performance of ConvNeXt is further compared with other deep learning models, including ResNet-50, DenseNet-121, MobileNetV3, EfficientNet-B3, and VGG-16. Experimental results demonstrate that ConvNeXt outperforms these models, offering a lightweight yet highly efficient solution for papaya leaf disease detection. To enhance transparency, the Grad-CAM technique is utilized to visualize disease-affected regions such as lesions and discoloration, ensuring model interpretability. The Llama 3 8B LLM advisory module improves interpretability and provides Treatment recommendation in actual agricultural disease surveillance scenarios. The model is trained on a dataset comprising healthy and diseased papaya leaves and achieves an accuracy of 96%. Overall, the proposed system enables early detection, reliable analysis, and effective disease management, contributing to precision agriculture and supporting farmers in informed decision-making.

## **Keywords:**

Papaya Leaf Disease, ConvNext, Disease detection, Gradcam, LLM ,Treatment Recommendation

## **I.Introduction:**

Agriculture continues to be the backbone of the Indian economy. The role of plant diseases in the economy is significant in respect to the quantity and quality of crops produced. In India,

major Papaya growing states are Andhra Pradesh, Gujarat, Maharashtra, and Tamil Nadu. Among these, Papaya cultivation in Andhra Pradesh always tops the list as the major producer in respect to favorable climatic conditions and large scale cultivation. In particular, tropical crops such as Papaya are affected by plant diseases. The early detection of plant diseases in leaves plays an important role in controlling the widespread destruction of crops. Among tropical fruit crops, Papaya holds a special place in respect to nutritional, medicinal, and commercial value. Papaya leaf diseases are Anthracnose, Bacterial Spot, Curl, Healthy, Mealybug, Mite Disease, Mosaic, Ringspot. These Papaya leaf diseases occur in the form of curling, yellowing, and the appearance of white powdery patches, necrotic spots, and discoloration. The paper presents a comprehensive set of images of papaya leaves with various disease classes and healthy leaves. This dataset contains various variations in lighting conditions, orientation of leaves, and background, which makes this dataset highly suitable for real-world agricultural conditions[23]. These diseases are usually caused by various viruses, fungi, and bacteria, and are often influenced by various environmental factors such as humidity, temperature, and rainfall. If these diseases are not diagnosed early, they often spread very fast, resulting in huge losses for farmers in terms of their income. Traditionally, to detect diseases in papaya leaves, manual inspection by agricultural experts is carried out, which is usually time-consuming and often based on human error. However, with the advent of Deep Learning in recent years, various algorithms such as Convolutional Neural Networks (CNN) and other algorithms such as densenet-121 and modilenet-v3 have been applied for accurate classification of diseases in papaya leaves. The results of their study proved that transformer-based models perform better than existing conventional CNN-based models in terms of classification accuracy, robustness, and generalization ability in different environmental conditions. This proves the shift from conventional deep learning models to hybrid models in precision agriculture applications [15].

Despite achieving a high classification accuracy, deep learning models face another major drawback: a lack of interpretability. This makes deep learning models unreliable in agricultural applications. To address this issue, a new set of Explainable Artificial Intelligence (XAI) techniques has been developed. Among all the techniques, Grad-CAM (Gradient Weighted Class Activation Mapping) is one of the most commonly used techniques. This technique helps in visualizing the regions of an image that are contributing more to the classification results. By visualizing regions such as lesions and infected regions, Grad-CAM helps in building trust among users. Moreover, recent developments in Large Language Models (LLMs) have opened a new gateway to incorporating intelligent recommendation systems in precision agriculture applications. LLM helps in generating explanations based on detected diseases. This integration would fill the gap between disease detection and useful information, helping farmers obtain precise information regarding disease management, prevention, and treatment.

The major contributions of this paper are as follows:

1. Papaya leaf diseases such as Papaya Leaf Curl, Anthracnose, and Mosaic Virus mainly affect papaya leaves. These diseases reduce photosynthesis, plant vigor, and overall yield and quality. They also affect fruit quality.
2. The paper proposes a novel framework using ConvNeXt-Tiny and LLaMA 3 8B1 LLM for automated papaya leaf disease detection, explanation, and treatment recommendation.
3. The paper proposes a novel optimized ConvNeXt-Tiny disease classification model with 96% validation accuracy and low computational complexity.

4. The paper proposes a knowledge-driven LLaMA-3 8B LLM for generating context-aware disease explanations and organic/chemical treatment recommendations.

The remaining part of the paper is organized in various sections. In the section II explains the 'Related work,' a literature survey of the previous works done on the previous methods has been included. In the section III Materials and method shows the 'Dataset Description,' the techniques used in the collection, augmentation, and pre-processing of the Dataset have been discussed. In the section IV 'Proposed Methodology,' has been discussed. In the section V 'Evaluation Metrics,' has been discussed. In the section VI 'Grad-CAM Based Visual Interpretation,' has been discussed. In the section VII 'Result and Discussion,' work has been included. In the section VIII 'Conclusion,' the future scope of the work has been included.

## **II.Related work:**

The papaya plant is very vulnerable to various diseases that mainly target the leaves, resulting in a substantial decrease in the productivity of the crops. Among the diseases, papaya ringspot virus, leaf curl disease, and anthracnose are the most frequently observed ones that cause substantial damage to papaya plantations. In recent years, vision transformer models, multimodal learning techniques, and lightweight deep learning models have been considered to improve plant disease detection systems. The main aim of these techniques is to improve the accuracy of the classification model and make it feasible for implementation in a smart agriculture system. Huang et al. this study proposed a model for apple leaf disease classification proposed a fusion model that combined ConvNeXt and transformer models to extract local and global features. The proposed model has excellent generalization capabilities and can improve the accuracy of classification in identifying patterns in apple leaves [1]. Roumeliotis et al. shows the Multimodal models that combine CNN and large language models have been proposed for use in general plant disease detection. [2]. Oad et al. proposed the Ensemble learning methods and explainable artificial intelligence have been suggested for plant leaf disease identification in different crops. These methods combine multiple classifiers to enhance the accuracy of predictions. Visualization methods also emphasize key disease characteristics in leaf images, making the prediction model more transparent [3]. Karimanzira et al. Context-aware deep learning models have been suggested for disease identification in tomato plants by considering environmental and contextual factors. These models examine leaf images and contextual factors to enhance disease identification accuracy in practical agricultural settings. These models can also address challenges related to lighting, background, and agricultural settings [4]. Salmon et al. Vision transformer models and mixture-of-experts learning strategies have been suggested for different plant species in agricultural settings. These models can adaptively assign expert subnetworks for examining different disease patterns on plant leaves. The adaptive model can enhance accuracy in agricultural datasets [5]. Ergun et al. Hybrid models that include ConvNeXt and vision transformer models have been used in the detection of diseases in mango leaves. [6]. Kaur et al. Visual language models that include the use of a transformer have been used in the detection of diseases in tomato leaves in greenhouses. The models have the ability to integrate the semantic interpretation of text and the extraction of visual features. The models have demonstrated the ability to develop AI-based models that can be used in the detection of diseases in plants [7]. Sujatha et al. Several studies have been conducted to determine the integration of machine learning and deep learning models in the detection of diseases in plants. [8]. Mustofa et al. Lightweight ensemble learning algorithms have been proposed for disease classification in the leaves of the soursop plant. These algorithms use multiple classifiers to enable efficient and accurate disease classification with low computational complexity. Explainable AI algorithms are also used to emphasize the leaf

areas that contribute to disease classification [9]. Tahsin et al. Semi-supervised deep learning algorithms have been used for disease detection in betel leaf plant species using pre-trained models and explainable AI algorithms. These algorithms are very effective when the amount of labeled training data is small. The combination of transfer learning and explainability is very effective in improving the accuracy and interpretability of plant disease detection systems [10]. Nahiduzzaman et al. Explainable deep learning algorithms have been proposed for automatic disease classification in mulberry plant leaves. Visualization algorithms are used to emphasize the areas of the leaves that are diseased, making it easier for users to understand the disease classification results. These explainability algorithms make AI-based agricultural diagnostic systems more trustworthy and transparent [11]. Mengisti et al. Explainable AI, in association with knowledge distillation techniques, has been used for medicinal plant leaf classification. This helps in developing efficient models that have similar performance characteristics to complex deep learning models but with reduced computational needs. The efficiency of models is important in real-world applications in agriculture, where devices may have limited computational capabilities [12]. Nigar et al. Deep learning-based prediction models in association with explainable AI have been used for plant leaf disease classification functions. These models help in accurate predictions and provide a clear idea about the explanation of the diseased region in the plant leaves. This association with deep learning models helps in better understanding of the decision-making process by users [13]. Hossain et al. Vision transformer models have been used for medicinal plant leaf classification using deep learning-based image analysis techniques. These models are highly efficient in capturing complex patterns and structural features in plant leaves, which are used for plant species identification and classification functions [14]. Parmar et al. Transformer-based architectures have also been applied for disease identification in papaya plants, where leaf images are analyzed using vision transformer models. These types of models are very effective in identifying disease patterns in papaya leaves. [15]. Borhani et al. employed vision transformer models for the automatic classification of plant diseases, and the performance was outstanding. However, the model is computationally expensive and requires high-performance GPUs. Moreover, the model's performance is poor when the dataset is small, and there is a problem of overfitting [16]. Kalpana, et al. introduced Swin transformer models with residual convolutional layers for disease detection. The model is complex and computationally expensive, so it is not suitable for implementation in portable devices. Moreover, the model's performance is poor when the dataset is imbalanced, and there is a need for better data augmentation techniques [17]. Selvaraj, R. et al. proposed an attention-based vision transformer optimized by using hybrid algorithms for turmeric leaf disease identification. Though accuracy is improved by using optimization, this method is faced with problems such as hyperparameter tuning, which is difficult. In addition, this method is not applicable in areas with limited data sets due to the use of large data sets in training the model. Mulugeta, et al. conducted a systematic review on the use of deep learning in classifying medicinal plants. From this systematic review, there are promising developments in this area. However, this systematic review also revealed that there is a lack of diversity in data sets, with most being done on a few species. In addition, there is no standardization in benchmarks, which makes it difficult to compare results. This makes progress in this area difficult [19]. Kavitha, et al. offered a real-time deep learning solution for the identification of medicinal plants. Although the solution is effective, it has limitations in terms of training data availability and susceptibility to lighting conditions. The proposed solution's dependency on controlled datasets reduces its generalizability to real-world agricultural environments [20]. Bhoyan et al. M. F. offered a dual-attention guided deep learning solution with interpretability. Although the proposed solution is more interpretable, its dual-attention mechanism incurs high computational complexity, which is a limitation for its application in low-power devices. The proposed

solution also demands large-scale training datasets, which are not readily available for small-scale farmers [21]. M. Rahman, et al. offered deep learning and vision transformers for the classification of medicinal plant leaves. Although the proposed solution is effective, it has limitations in terms of small-scale datasets and lacks interpretability, which makes it difficult for end-users to trust the solution. The proposed solution is also susceptible to environmental noise, which further hampers the performance of the solution in uncontrolled agricultural field settings [22]. Wiwik et al. introduced an optimized CNN ensemble learning method for herbal leaf disease identification. Although the proposed ensemble learning method is robust, it has redundancy and high complexity. The proposed method also has scalability problems since training multiple models simultaneously is computationally expensive [23]. Hasan A., et al. introduced an explainable deep learning model integrating EfficientNetB7-MHA with zero-shot segmentation. Although the proposed model is innovative, it requires a large number of labeled images and is not effective for new classes of diseases. The complexity of the proposed model further makes it inappropriate for real-time applications in agricultural settings with resource constraints [24]. Ahmed et al. employed ResNet-101 for guava disease classification using transfer learning. Although transfer learning is more efficient, there are chances of overfitting since the model is applied to small domain-specific datasets. Moreover, the model is not explainable, and it is difficult for farmers to understand the outputs [25]. Mustofa et al. introduced BDPapayaLeaf, a dataset for papaya leaf disease detection. Although it is a good dataset, it is geographically limited and lacks diversity, making it hard to generalize over agricultural environments around the world. In addition, the dataset is still considered small compared to other large-scale datasets [26]. Banarase, et al. introduced a fusion of CNN and transfer learning fusion for papaya leaf disease identification. Although the fusion approach improved accuracy, the model is not accurate for new disease classes and is highly reliant on image preprocessing [27]. Zhang et al. developed a light-weight model for rice leaf disease identification. Although it is light-weight, the model has lower accuracy in complex agricultural environments with overlapping symptoms of diseases. The model also struggles to adapt to different climatic conditions [28]. Medikonduru et al. proposed a diversity-optimized deep ensemble approach for plant leaf disease classification. Although robust, the ensemble approach has high latency during inference. The approach also faces challenges in scaling up for large agricultural datasets [29]. Tuteja et al. proposed CNN Att LSTM for jackfruit leaf disease classification. Although efficient, the hybrid CNN-LSTM approach has high risks of overfitting. The approach also requires large amounts of sequential data, making it unsuitable for small-scale agricultural environments [30]. Lekkala et al. proposed ConvNeXt-Small for papaya leaf disease classification. Although efficient, the shallow approach has less capability to extract complex disease features. The approach also faces challenges in generalizing to different agricultural datasets [31]. Kılıç et al. examined the classification of papaya leaves via multi-class and binary classification using Darknet models. Although optimized, the models are still vulnerable to class imbalance and heavy preprocessing. Their reliance on high-quality images impacts their performance in noisy real-world environments [32]. Gulzar et al. performed a comparative analysis of pre-trained deep learning models for papaya leaf disease classification. Although transfer learning speeds up the training process, the models are not domain-specifically fine-tuned. This drawback affects the model's performance when deployed on agricultural datasets with distinct disease patterns [33]. Ahmed et al. proposed a deep learning approach for papaya leaf disease classification. Although the approach is successful, it is not interpretable and fails to adapt to real-time environments. Its dependence on large datasets also affects its applicability in resource-poor agricultural settings [34].

The existing plant and medicinal leaf disease classification approaches also have some limitations in their real-world applications. These limitations include high computational

complexity, dependence on large datasets, overfitting, poor generalization performance on imbalanced data sets, and low adaptability to new classes. Moreover, the absence of standardized datasets, the impact of environmental factors such as lighting conditions, noise

images, and overlapping symptoms, and low model interpretability also limit the applicability of these approaches. To overcome these limitations, this paper presents a new vision-language agricultural intelligence approach based on ConvNeXt hierarchical feature extraction and LLaMA-3 8B. This approach uses large-kernel depthwise convolutional networks to improve the fine localization of plant diseases and uses the language model to derive interpretable predictions.

### III. MATERIALS AND METHODS

#### 3.1 Dataset collection

The papaya leaf dataset used during the course of this work has been carefully constructed by gathering information from multiple reliable open-access resources, combined with the acquisition of images at the field level to increase the diversity of the environment. The papaya dataset[36] consists of 1,500 high-resolution images per class, covering both healthy and diseased states under natural illumination conditions. Images were taken against plain backgrounds to avoid confusion regarding the morphological features, thereby improving the discriminability of the features by removing background noise. Images showing extreme levels of physical damage, deformity, occlusion, or illumination were avoided to maintain uniformity and avoid bias during the course of the work. All the images were resized to a uniform size of 224 x 224 pixels to match the specifications of the modern deep convolutional neural networks, including ConvNeXt architectures. The final dataset includes eight different classes of papaya leaves with various diseases like Anthracnose, Bacterial Spot, Curl, Healthy, Mealybug, Mite Disease, Mosaic, Ringspot, etc. The dataset is of multi-class type, which is suitable for the evaluation of the effectiveness of the learning of class-specific features. The images of the dataset are shown in Fig 1.

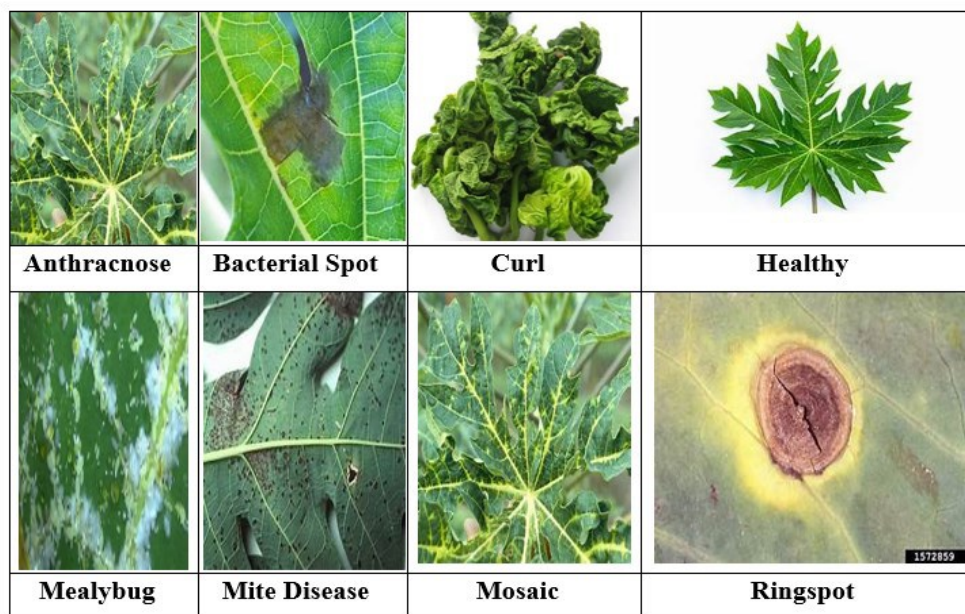


Fig 1: sample images of papaya Leaf

For robust experimental evaluation, the data set was divided into structured sets and 70% of the images were used for training, 15% for validation, and 15% for testing. The balanced classes and standardized preprocessing approach will improve the robustness of the model's training while avoiding overfitting and imbalance problems.

### 3.2 Image preprocessing and Augumentation

In addition to these basic normalization processes, various forms of data augmentation techniques were used to artificially enlarge the data set. These included geometric transformations such as rotation, horizontal and vertical flipping, zooming, cropping, and translation. And the images are shown in the Fig 2. These various forms of data augmentation techniques help to simulate various changes that can occur to a leaf under different conditions. This is helpful in improving the ability of the model to generalize well. Additionally, background homogeneity was ensured through the use of simple backgrounds that do not distract during image capture.



Fig 2: Data Augmentation Techniques Applied to Papaya Leaf Image (Horizontal Flip, Vertical Flip, Right Rotation, Left Rotation, and Brightness Enhancement)

To make the deep learning model more robust and have better generalization power, various data augmentation techniques were employed on the training data set. For example, random flip was employed to make the model sensitive to leaves regardless of their position on the image. Also, random rotation was employed to make the model insensitive to rotations of the leaves, which may occur when taking the image in the field. Additionally, color jittering was employed, which involves random changes to the brightness, contrast, and saturation of the image. This was done to make the model insensitive to changes in the light intensity of the colors of the leaves, which may result from the sun, shadows, or differences in camera capabilities. Finally, random cropping was employed to make the model focus on the most important part of the leaves by ensuring that the leaves are well-centered in the image.

From the below , Table I shows that the dataset was stratified before partitioning to ensure that there was proportional representation of each disease category in the entire set of experimental subsets.

TABLE 1.Dataset splitting for Disease Papaya Leaf

| Dataset Name | Class name | Total Images | Training Imges (70%) | Validation Images (15%) | Testing Images (15%) |
|--------------|------------|--------------|----------------------|-------------------------|----------------------|
|              |            |              |                      |                         |                      |



|                                                         |                |               |              |              |              |
|---------------------------------------------------------|----------------|---------------|--------------|--------------|--------------|
| Mendely<br>Dataset(Papaya<br>Leaf Disease<br>Detection) | Anthrachnose   | 1,500         | 1,050        | 225          | 225          |
|                                                         | Bacterial Spot | 1,500         | 1,050        | 225          | 225          |
|                                                         | Curl           | 1,500         | 1,050        | 225          | 225          |
|                                                         | Healthy        | 1,500         | 1,050        | 225          | 225          |
|                                                         | Mealybug       | 1,500         | 1,050        | 225          | 225          |
|                                                         | Mite Disease   | 1,500         | 1,050        | 225          | 225          |
|                                                         | Mosaic         | 1,500         | 1,050        | 225          | 225          |
|                                                         | Ringspot       | 1,500         | 1,050        | 225          | 225          |
|                                                         | <b>Total</b>   | <b>12,000</b> | <b>8,400</b> | <b>1,800</b> | <b>1,800</b> |

## IV.PROPOSED METHODOLOGY

### 4.1ConvNeXt

ConvNeXt (Convolutional Neural Network Next) is a modified version of the CNN architecture that aims to bridge the performance gap between traditional CNNs and Vision Transformers. The architecture maintains the hierarchical inductive biases of CNNs and incorporates architectural elements inspired by Vision Transformers, such as the use of large kernel sizes, layer normalization, inverted bottlenecks, and reduced architectural complexities.the overall architecture is shown in the fig 3.

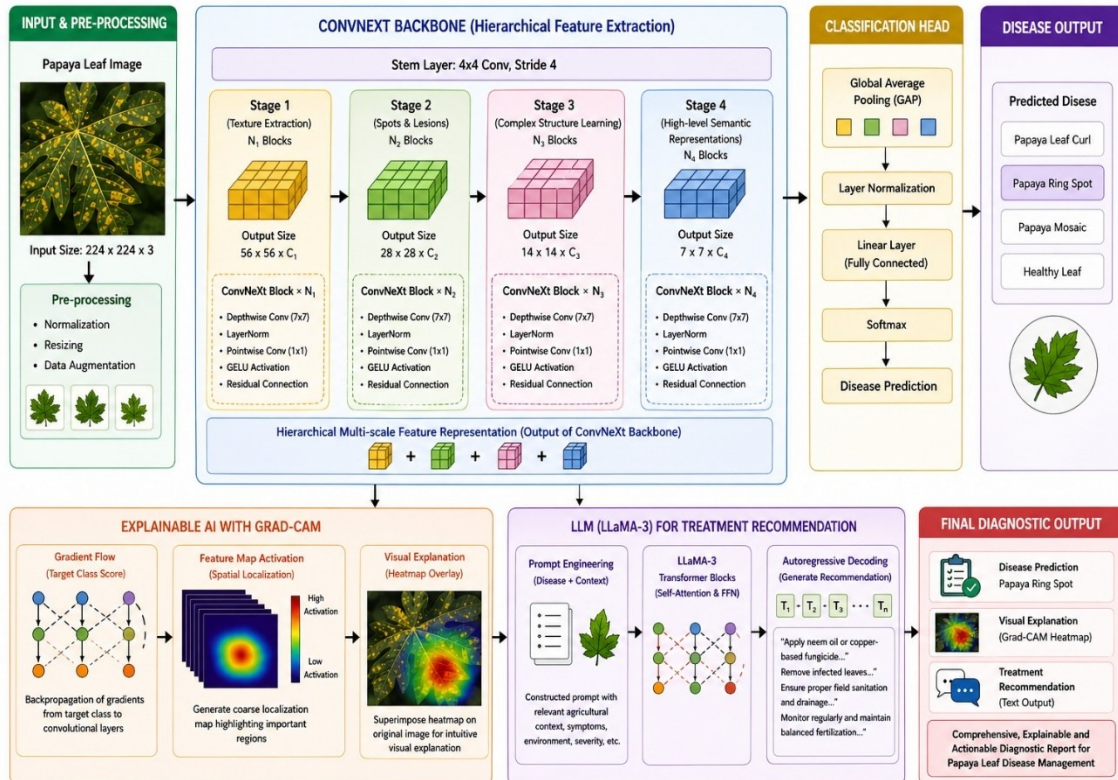




Fig 3: Overall Architecture of Papaya Leaf Disease Detection

In the proposed system, the ConvNeXt architecture has been employed as the backbone feature extractor for the classification of leaf images at high resolution. To begin with, the leaf images are subjected to various preprocessing techniques such as resizing, normalization, and augmentation for better generalization performance. The preprocessed leaf images are then fed into the stem layer of the ConvNeXt architecture, where the patchify convolution ( $4 \times 4$  convolution with stride 4) reduces the spatial resolution and increases the depth of the channels. This is followed by the ConvNeXt hierarchical structure. Each stage consists of the progressive extraction of more abstract semantic representations, which enable the model to focus on low-level texture information such as the veins on leaves or the colors, as well as high-level discriminative patterns associated with plant species or types of diseases.

The ConvNeXt architecture of the project has the following major layers and components:

- 1) Stem Layer – Patch Embedding Layer: This layer consists of a convolution operation to downsample the input image, which reduces the spatial dimensions of the input image to create feature maps through the extraction of structured feature tokens.
- 2) ConvNeXt Blocks: Each block consists of a  $7 \times 7$  depthwise convolution operation to extract spatial features, followed by a layer normalization operation to enable stable training. A  $1 \times 1$  pointwise path creates an inverted bottleneck to expand the channels before compressing the channels. Finally, the GELU activation operation provides non-linearity to the model, along with a residual connection to enable smooth gradient flow.
- 3) Downsampling Layers: These layers are placed between the stages to reduce the spatial dimensions and increase the number of channels to enable a hierarchical representation.
- 4) Global Average Pooling Layer - This layer reduces the spatial information into a concise vector form, hence reducing overfitting and computational complexities.
- 5) Fully Connected (Classification) Layer - This layer maps the learned feature representations to the class scores using the softmax activation function.

The hierarchical feature maps learned through the various ConvNeXt blocks are progressively refined to capture the subtle differences between medicinal leaves. The residual connections facilitate the smooth flow of gradients, and the large kernel depthwise convolutions provide spatial context, which plays a crucial role in detecting disease patterns and morphological differences.

Gradient Weighted Class Activation Mapping is a gradient-based method for providing spatial interpretability to CNNs. The method is used to explain the predictions of the CNN by determining the regions of the input image that contribute the most to the prediction of a specific class. The heatmap indicates high activation values over the regions where the diseases have manifested, such as discoloration of the leaves and damaged tissues, which indicates the effective utilization of the characteristics associated with plant diseases by the network.

The explanation module based on the Large Language Model (LLM) Llama 38B was tested to ensure its capability to produce interpretable disease diagnosis and management suggestions

from the predicted class outputs. After the classification model predicts a disease (for example, Anthracnose), the LLM examines the prediction and automatically produces short agronomic explanations for disease description, organic management practices, and suggested chemical controls with dosage.

## **4.2 State of Art Models**

### **a)DenseNet-121**

DenseNet-121 is a densely connected convolutional neural network with 121 layers, which improves feature propagation through the network by connecting each layer with all the following layers in a dense block. In contrast to traditional CNNs, which use sequential layer connections, DenseNet-121 utilizes concatenation of feature maps, which improves gradients as well as discriminative learning. The network has an initial layer of convolution followed by a max pooling layer, followed by four dense blocks with transition layers between them, which are comprised of a convolution layer with a kernel of 1x1 followed by an average pooling layer. In this research, the DenseNet-121 network has been employed as a baseline network to check the efficacy of the proposed ConvNeXt-based papaya leaf disease diagnosis system, with similar preprocessing steps employed for the images. The DenseNet-121 network has a number of advantages, including parameter efficiency, propagation of information, as well as reusing features, but it has a number of disadvantages, including memory usage, a limited receptive field, as well as a loss of information during the transition layer.

### **b) EfficientNet-B3**

EfficientNet-B3 is a compound scaling convolutional neural network architecture that aims to achieve accuracy while minimizing computational complexity. It is based on an architecture consisting of an MBConv block with squeeze and excitation module integration. The model balances scaling in depth, width, and resolution to achieve efficient and robust feature representation. Therefore, EfficientNet-B3 can be used in medical and agricultural image classification tasks. In this study, EfficientNet-B3 was used as a baseline model to validate the proposed system based on the ConvNeXt architecture for papaya leaf disease diagnosis. The proposed model uses a convolutional neural network with an initial convolutional stem followed by stacked layers consisting of an MBConv block with integration of depth separable convolution, batch normalization, and Swish activation functions for efficient feature extraction. In addition, progressive downsampling is used to extract characteristics such as lesions, discoloration, and texture changes in the papaya leaf images. Global average pooling and Softmax classification are then used to achieve compact feature representation for disease classification.

### **c) Mobilenet-v3**

MobileNetV3 is a lightweight and computationally efficient convolutional neural network architecture. It incorporates several components like depth-wise separable convolution, inverted residual mobile convolution (MBConv), squeeze and excitation attention, and hard Swish activation functions to achieve parameter efficiency while maintaining discriminative ability in feature learning. In this paper, MobileNetV3-Large (MNetV3-Large) is utilized as a baseline model to test the proposed papaya leaf disease detection system based on ConvNeXt architecture under similar conditions. The proposed

architecture begins with a convolutional stem to extract low-level features. Subsequently, a series of inverted residual mobile convolution (MBConv) blocks are utilized to efficiently extract spatial and channel features. The progressive downsampling helps in extracting disease-related features like lesions, chlorosis, mosaics, and vein distortion. Global average pooling and Softmax classification are utilized in the last stage for disease classification.

#### **d) ResNet-50**

ResNet-50 is a deep convolutional neural network with residual learning and skip connections to enable the efficient training of very deep models. The network has 50 layers, which are grouped into bottleneck residual blocks. The network is able to effectively learn hierarchical representations in complex image classification problems. In this study, the papaya leaves are preprocessed by resizing, normalization, and augmentation, after which they are fed into the network. The network starts with a convolutional layer and a max pooling layer to extract low-level image features, which include edges, gradients, etc. The network is followed by four residual blocks, each containing bottleneck blocks with 1x1, 3x3, and 1x1 convolutional operations to reduce dimensions, extract spatial features, and increase channels, respectively. The residual connections enable deeper feature learning while maintaining important disease-specific patterns, which include lesions, chlorosis, mosaics, and vein distortions. Global average pooling is used, followed by a fully connected layer with Softmax activation to classify the image.

#### **e) VGG-16**

VGG-16 is a deep CNN with 13 convolutional layers and 3 fully connected layers, which uses uniform 3x3 kernels and 2x2 max pooling for hierarchical feature extraction. It has the ability to learn low-level features like edges, mid-level features like textures, and high-level features like semantics, which are useful for discriminating between slight variations of diseases. VGG-16 has been used as a transfer learning model for the classification of diseases of the papaya leaf. The input images of different diseases are first normalized by resizing them to a size of 224x224, followed by the application of ImageNet normalization. The images are then fed into a series of convolutional layers with stacked convolutional blocks, which have the ability to learn features like lesions, mosaics, discoloration, and vein distortions of diseases, followed by max pooling layers that reduce the dimension of the images. The learned features of the images are then fed into fully connected layers, which replace the original classifier with a new classifier for the classification of the diseases into eight different classes, followed by a Softmax classifier.

### **V. Evaluation Metrics**

The definition of the papaya leaf disease classification framework is given as a multi-class classification problem on the images, where there are  $K$  classes of diseases, which include the healthy state as well as the diseased state. The performance of the classification model is given by the confusion matrix, which is derived based on the predicted classes and the actual classes of the test data. The definition of the confusion matrix for each class  $i$  is given as True Positives ( $TP_i$ ), False Positives ( $FP_i$ ), False Negatives ( $FN_i$ ), and True Negatives ( $TN_i$ ). In the case of the papaya leaf disease classification problem, the True Positives are the correctly classified images of class  $i$ , the False Positives are the images of the healthy state or the other classes of diseases incorrectly classified as class  $i$ , and the False Negatives are the instances of the disease.

$$\text{Accuracy} = \frac{\sum_{i=1}^K TP_i}{\sum_{i=1}^K (TP_i + FP_i + FN_i)}$$

$$\text{Precision}_i = \frac{TP_i}{TP_i + FP_i}$$

$$\text{Recall}_i = \frac{TP_i}{TP_i + FN_i}$$

$$F1_i = 2 \cdot \frac{\text{Precision}_i \cdot \text{Recall}_i}{\text{Precision}_i + \text{Recall}_i}$$

$$(\text{Macro Average} = \frac{1}{K} \sum_{i=1}^K M_i)$$

$$\text{Weighted Average} = \frac{\sum_{i=1}^K n_i M_i}{\sum_{i=1}^K n_i}$$

$$\mathcal{L} = - \sum_{i=1}^K y_i \log(\hat{y}_i)$$

$$\text{AUC} = \int_0^1 \text{TPR}(\text{FPR}) \, d(\text{FPR})$$

### 5.1 Training and Validation Accuracy Curves of ConvNext and Comparision of state of the Art models :

The training and validation accuracy curves, as depicted in Figure X, were used to investigate the learning behavior and generalization ability of the proposed deep learning models for papaya leaf disease classification. The proposed ConvNeXt model has demonstrated a stable and efficient learning process, in which the training accuracy has rapidly increased during the initial epochs and then converged to the optimal value. Meanwhile, the validation accuracy curve has shown a similar pattern to that of the training accuracy, indicating that the proposed model has a low risk of overfitting and thus a high ability to generalize. As can be seen, the proposed ConvNeXt model has achieved a high overall classification accuracy of 97-98%, indicating that the proposed model can efficiently learn discriminative features

from the papaya leaf images and thus classify the healthy and diseased classes. Similarly, for the DenseNet121 model, a smooth learning curve is observed, with training and validation accuracy converging to 96-97%. This is due to the strong feature reuse and gradient propagation. The EfficientNet-B3 model also performed well in terms of validation accuracy, with convergence to 95-96% for both training and validation. This is due to the good balance between complexity and accuracy for this model. However, for the MobileNetV3 model, slightly lower validation accuracy is observed, ranging from 92-94%, which is due to the lightweight design of this model for mobile and edge devices. Similarly, for the ResNet-50 model, moderate performance is observed, with validation accuracy ranging from 90-93%, indicating limited ability to learn complex disease patterns. The VGG-16 model is observed to have a smooth learning curve but takes more epochs to converge due to the complexity of this model.

In general, the accuracy analysis above confirms that the proposed model based on ConvNeXt has better learning stability and higher accuracy in classification compared to the conventional convolutional neural network model. The fact that the training and validation curves are very close further indicates the robustness of the proposed framework.

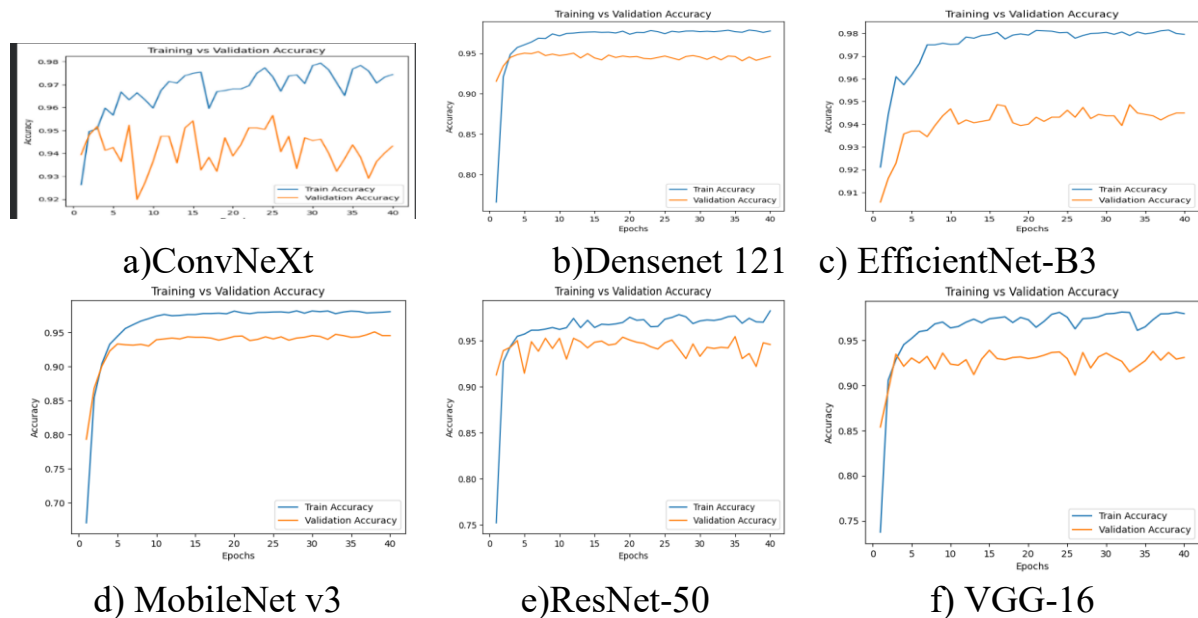


Fig 4: Accuracy Curves of Convnext and Comparison Models

## 5.2 Training and Validation Loss curves of ConvNext and Comparison of state of the Art models :

The total loss curves of the training and validation results of the tested models are shown in Figure X. By examining the trends of the loss curves, it can be seen that the proposed ConvNeXt model exhibits a stable and consistent learning pattern throughout the training process. It can be seen that the training loss continuously decreases over the training process, while the validation loss remains stable without major fluctuations. This phenomenon proves that the ConvNeXt model successfully extracts discriminative features from the papaya leaf dataset and exhibits good generalization ability without major overfitting issues. Moreover, the

smooth convergence pattern also implies that the model learns good representations of the disease-related features, including texture changes, lesion structures, and discoloration patterns in papaya leaves. Likewise, the DenseNet-121 model presents a fast drop in the training loss in the early epochs, followed by a stable phase as the training process continues. The validation loss also decreases but with some fluctuations in the latter epochs, which reveals that while the model succeeds in learning effective hierarchical representations, it may suffer from mild overfitting as the training process continues. On the other hand, the EfficientNet-B3 model presents fast convergence in both training and validation losses, with both losses dropping sharply in the early training epochs. However, the validation loss becomes stable at a relatively higher value compared to the training loss, which suggests that the model succeeds in learning the dominant patterns but may not generalize well to new examples. The light MobileNetV3 model presents a steady drop in the training loss, but the validation loss varies significantly during the training epochs. The same phenomenon can be observed in ResNet-50, in which the loss value tends to reduce over time, while the validation loss varies in an irregular manner. This shows that the model is moderately unstable in terms of generalization. In addition, the convergence speed of VGG-16 is slower compared to the other architectures, as indicated by higher validation loss and fluctuations in the loss value during the training process. This shows that the model is facing difficulties in learning the complex disease patterns in the papaya leaf dataset.

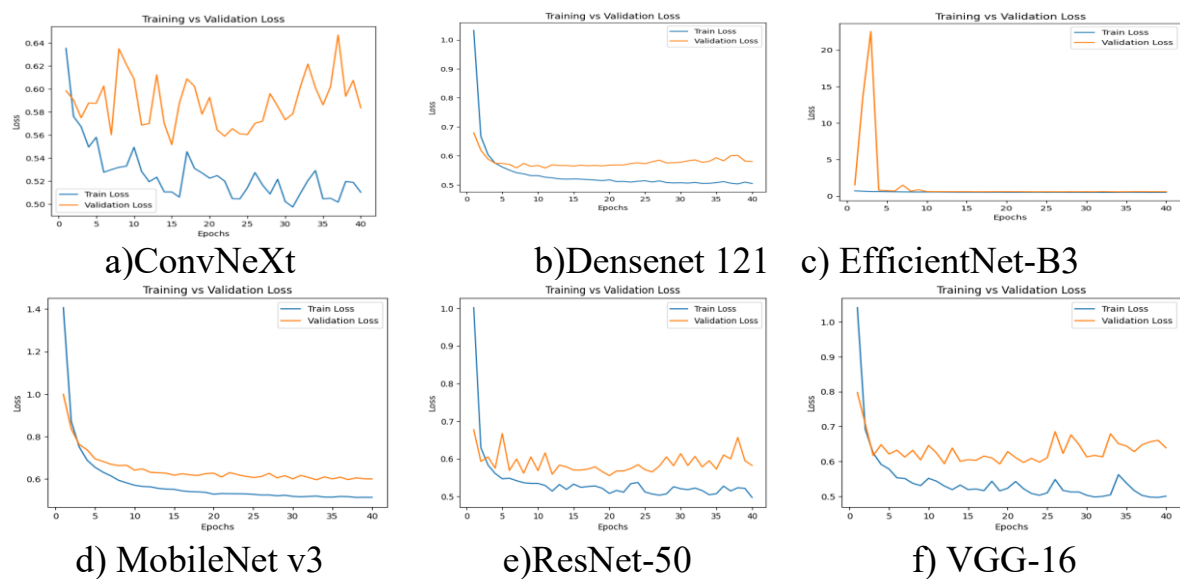


Fig 5: Training and Validation curves of Convnext and Comparison Models

### 5.3 Roc of ConvNext and Comparision of state of the Art models :

The ROC curves and AUC values shown in Figure X were used to assess the classification ability and sensitivity of the proposed models to classify the papaya leaf disease classes. The proposed ConvNeXt model outperformed all other models in terms of discriminative ability for all classes. The model performed close to perfectly with AUC = 0.99 for the Healthy class, AUC = 0.99 for the Papaya Ring Spot Virus class, and AUC = 1.00 for the Anthracnose class, which indicates a very high ability to distinguish between healthy and diseased leaf images. Similarly, the DenseNet121 model performed well in terms of discriminative ability with AUC



values close to 0.99 for most classes, which confirms its high robustness in identifying disease-specific visual features. The Efficient Net B3 model also performed well with competitive AUC values ranging between 0.96 and 0.98, which indicates high classification performance with high efficiency. In contrast, the MobileNetV3 model performed moderately with AUC values ranging between 0.93 and 0.96, which indicates its high efficiency but slightly lower discriminative ability compared to other models. The performance of the ResNet50 model has been relatively lower, with AUC values ranging between 0.88 and 0.92, indicating some difficulties in differentiating visually similar patterns of diseases. From the overall analysis of the ROC curve, it is clear that the proposed framework, which is based on the ConvNeXt model, has better class separability and can differentiate diseases more efficiently compared to other conventional CNN models. The higher values of AUC, which have been achieved in the case of all the diseases, indicate the efficiency of the proposed approach in identifying papaya leaf diseases.

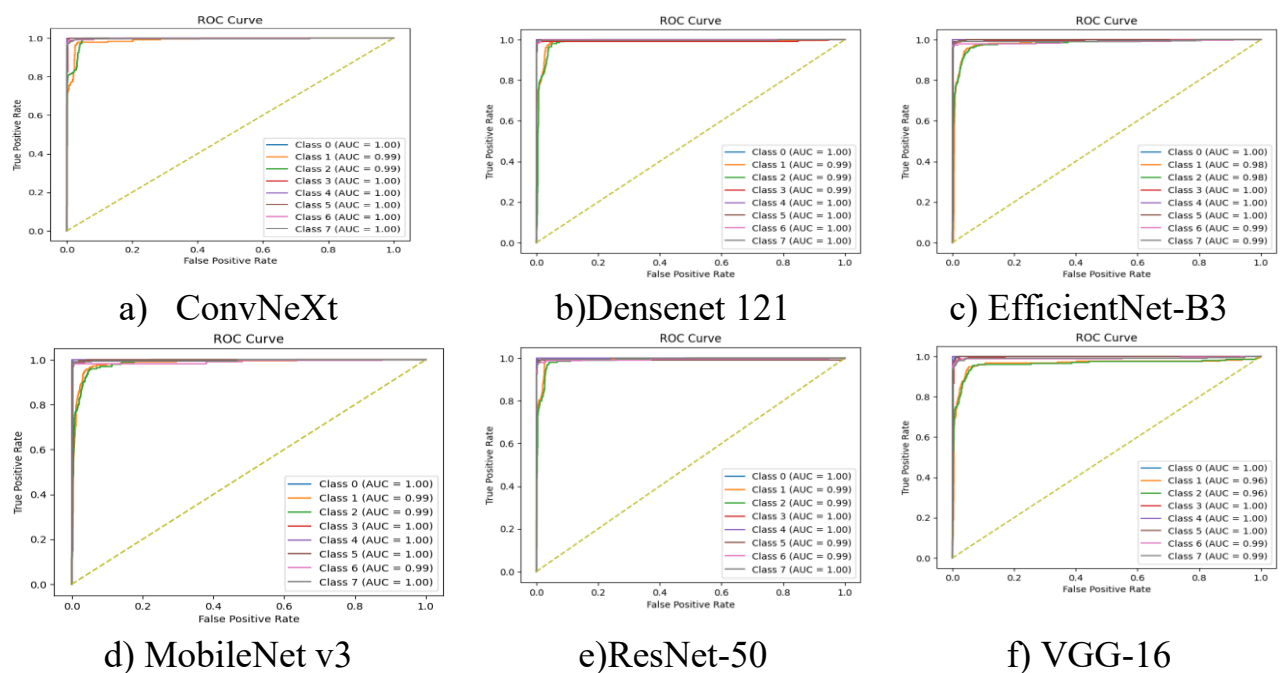
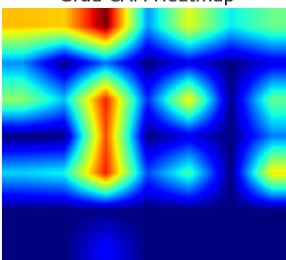
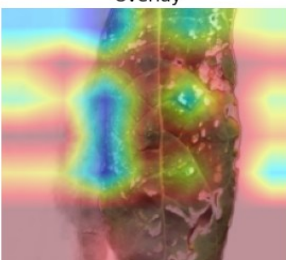
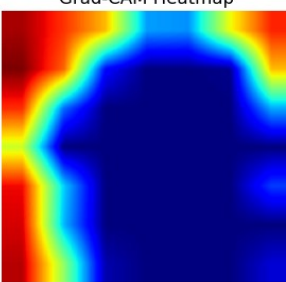
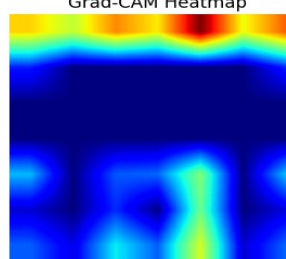
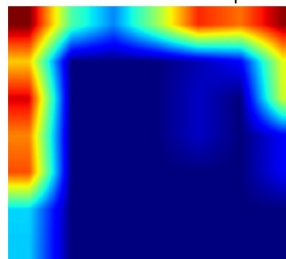
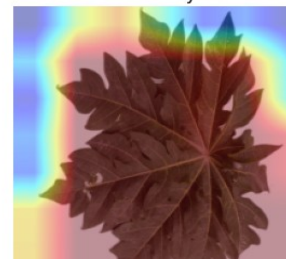
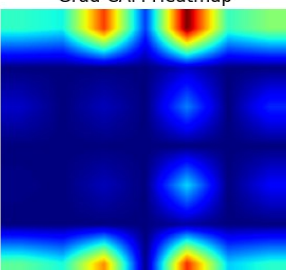


Fig 6: ROC Curves of Convnext and Comparison Models

## VI. Experimental Results:

### 6.1 ConvNeXt Model

The importance weights are then used to compute a coarse localization map of the regions of the input image that contribute to the prediction of the class. In the diagnosis of diseases in the field of agriculture, it is important to ensure that the predictions of the CNN are based on the symptoms of the leaves rather than the backgrounds. From the visualization results, it can be concluded that the proposed ConvNeXt architecture has generated a highly concentrated activation map, which is perfectly aligned with the infected regions of the leaf.

| Disease        |                                                                                                                                         | Gradcam-Heatmap                                                                                                                             |                                                                                                                                      |
|----------------|-----------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------|
| Anthracnose    | <p data-bbox="494 246 638 268">Original Image</p>      | <p data-bbox="813 246 1021 268">Grad-CAM Heatmap</p>      | <p data-bbox="1228 246 1340 268">Overlay</p>      |
| Bacterial Spot | <p data-bbox="494 560 638 582">Original Image</p>      | <p data-bbox="813 560 1021 582">Grad-CAM Heatmap</p>      | <p data-bbox="1228 560 1340 582">Overlay</p>      |
| Curl           | <p data-bbox="494 884 638 907">Original Image</p>     | <p data-bbox="813 884 1021 907">Grad-CAM Heatmap</p>     | <p data-bbox="1228 884 1340 907">Overlay</p>     |
| Healthy        | <p data-bbox="494 1198 638 1220">Original Image</p>  | <p data-bbox="813 1198 1021 1220">Grad-CAM Heatmap</p>  | <p data-bbox="1228 1198 1340 1220">Overlay</p>  |
| Mealybug       | <p data-bbox="494 1512 638 1534">Original Image</p>  | <p data-bbox="813 1512 1021 1534">Grad-CAM Heatmap</p>  | <p data-bbox="1228 1512 1340 1534">Overlay</p>  |

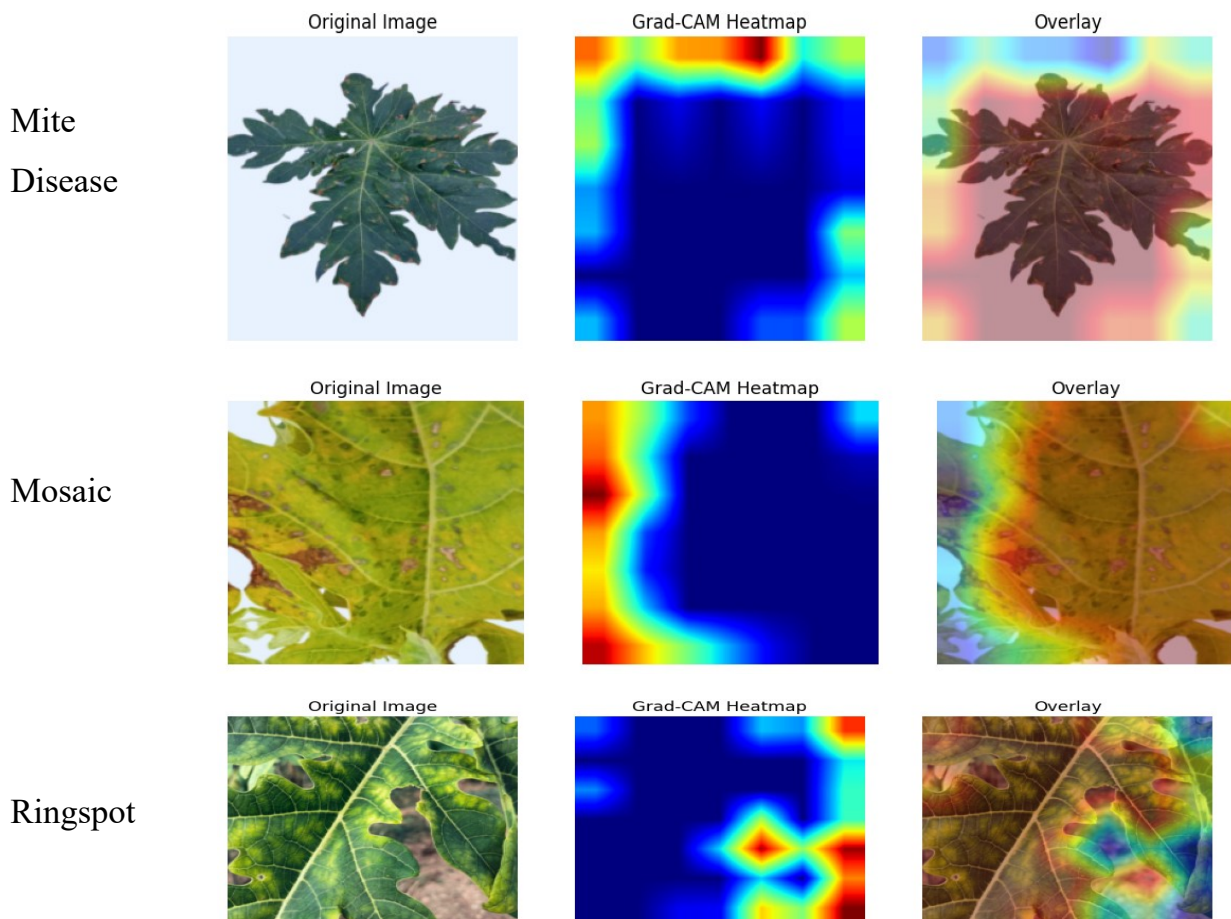
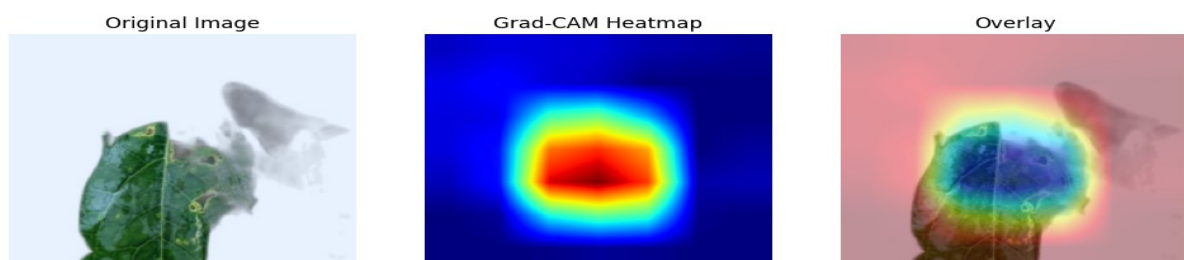


Fig 7: Grad-Cam Analysis Of Convnext model

## 6.2 Comparison of grad-cam of the state of the models:

The Grad-CAM visualizations show that there are considerable differences in feature localization across all comparative models. As shown in Figure 9, DenseNet-121 has partial attention to diseased regions; however, this attention is diffused over the regions, showing that this model has limited precision in feature localization. EfficientNet-B3 has multiple scattered zones of activation, showing attention to broader texture features rather than strict attention to regions that are specific to infection. MobileNet-V3 shows low intensity with uniform activation, suggesting that this model has limited capability to capture fine features of diseases. On the other hand, ResNet-50 shows moderate feature localization; however, this localization is diffused over regions, showing that this model has limited specificity.



b)Densenet 121



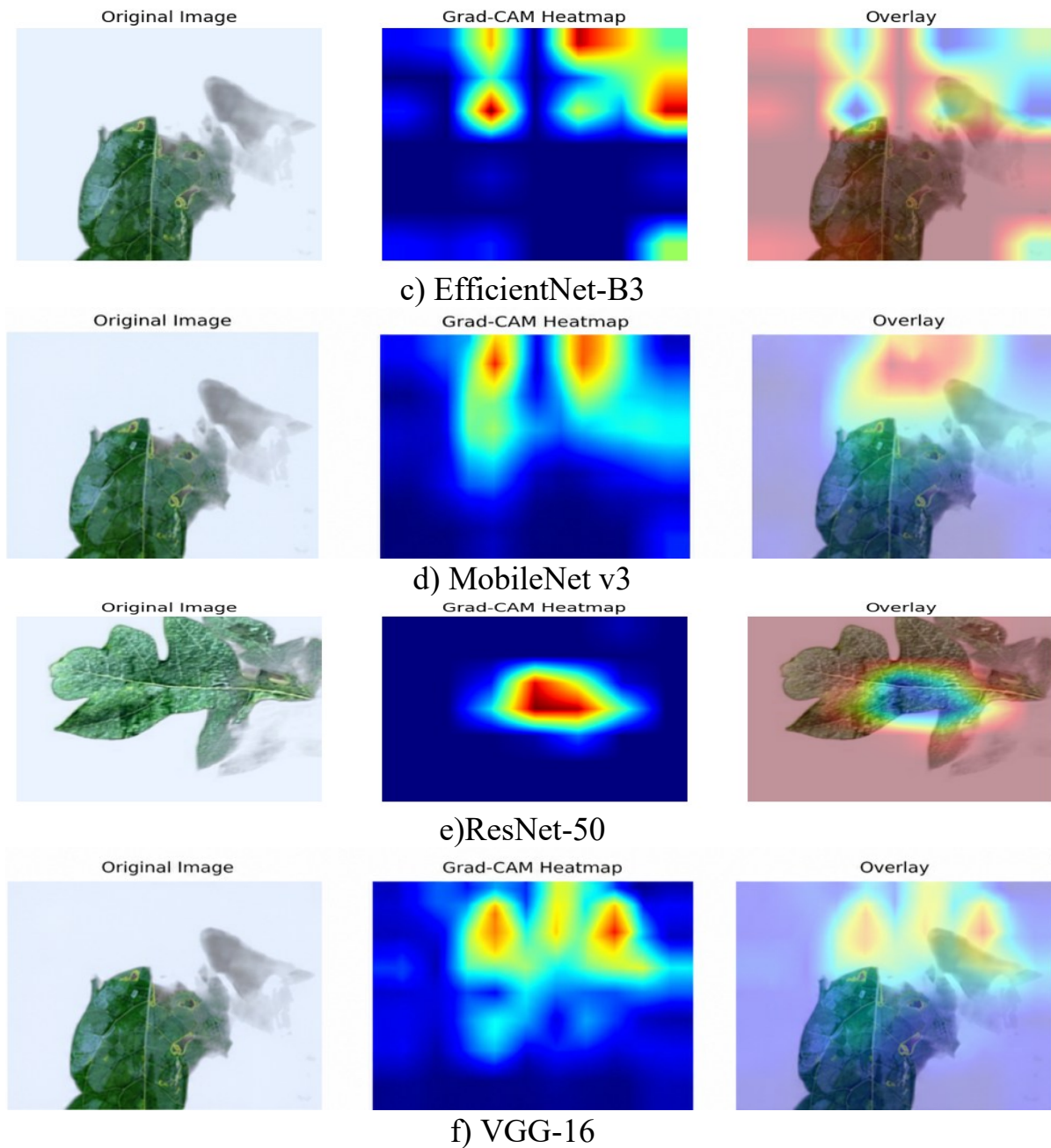


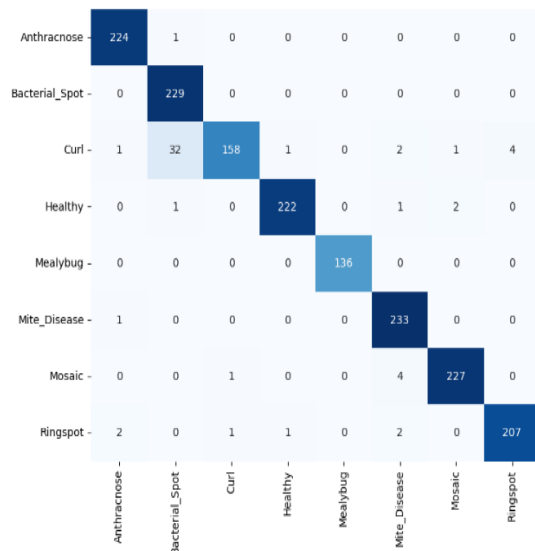
Fig 8: Grad-Cam Analysis Of Comparison Models

VGG-16 shows diffused activation, suggesting that this model has limited focus on regions affected by diseases. Overall, as shown across all models, ConvNeXt has better feature localization with concentrated feature activation, showing that this model has better capability to capture fine features of diseases, thereby improving interpretability for papaya leaf disease classification.

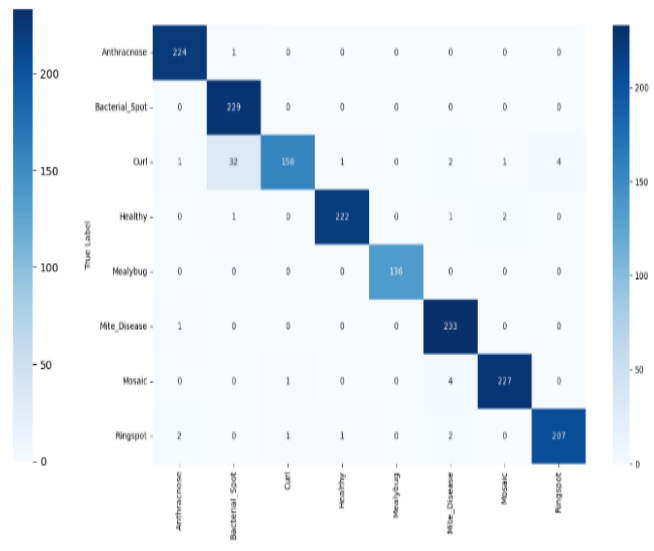
### 6.3 Confusion Matrix of ConvNeXt and Comparison of state of the Art models :

In this study, confusion matrices are computed for the proposed ConvNeXt-based model and comparison models such as DenseNet121, MobileNetV3, ResNet-50, EfficientNet-B3, and VGG-19. The proposed ConvNeXt model presents a strongly dominant diagonal distribution

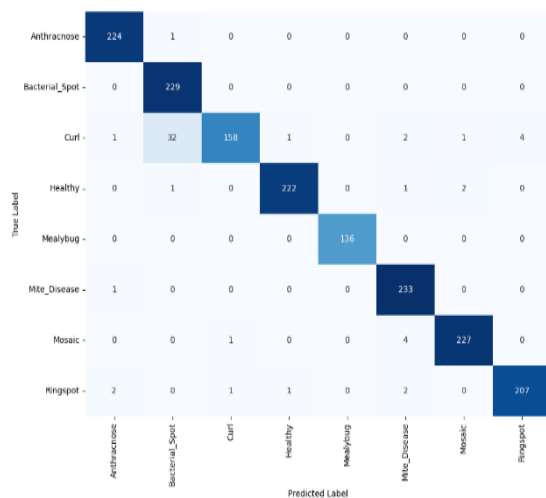
with minimal off-diagonal dispersion, reflecting high class separability and low class ambiguity. Conversely, DenseNet121 and EfficientNet-B3 models present moderate levels of confusion between classes of structurally similar diseases, reflecting slightly diffused contextual feature aggregation. The ResNet-50 model shows some instances of misclassification between classes of early-stage and advanced-stage symptoms, possibly due to residual feature blending in different stages of convolution. The lightweight MobileNetV3 model presents higher levels of off-diagonal elements in complex classes of diseases, reflecting limitations in compact feature representation. Similarly, the VGG-19 model shows increased levels of confusion in texture-similar classes, reflecting its comparatively low capacity in contextual feature aggregation and discrimination. In terms of the reliability of diagnoses, the confusion matrix analysis is consistent with the superior discriminative learning capability of ConvNeXt in terms of learning the subtle variations of pathological classes. The lower intensity of the off-diagonal elements indicates improved feature localization and context modeling, which is consistent with the Grad-CAM analysis results. Additionally, the confusion matrix analysis allows for the determination of particular disease classes that are susceptible to confusion.



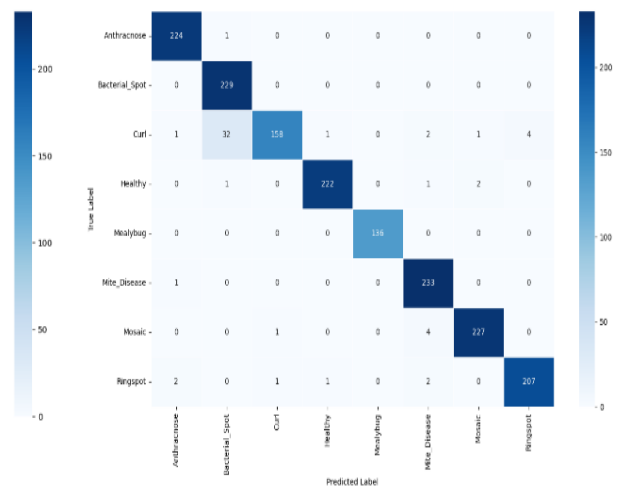
a) ConvNeXt



b)Densenet 121



c) EfficientNet-B3



d) MobileNet v3

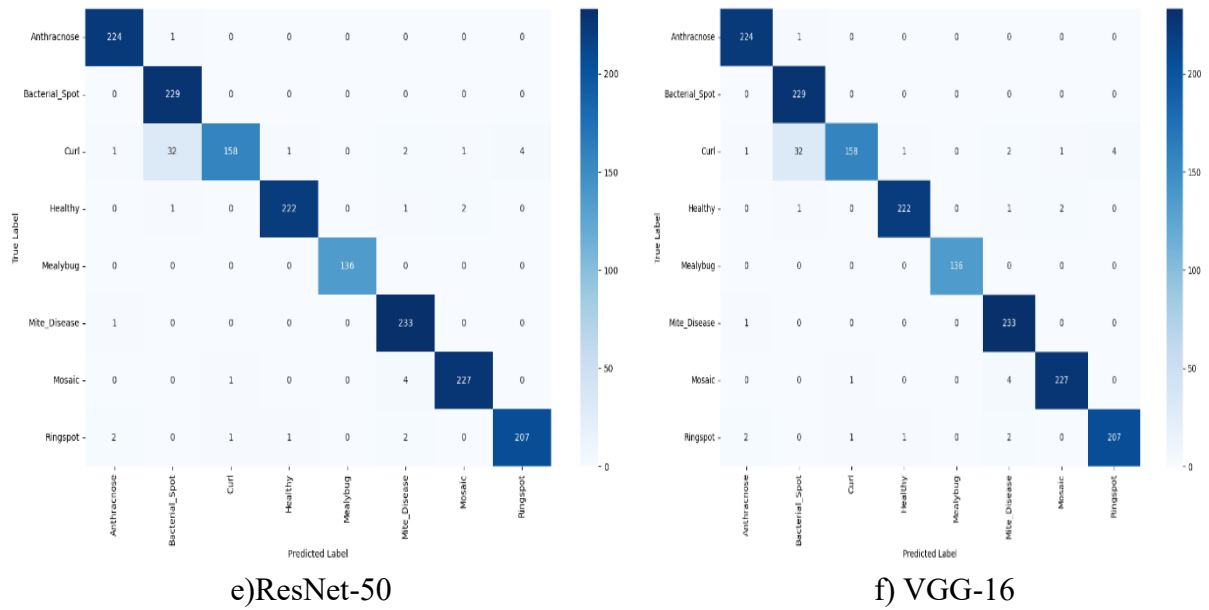


Fig 9: Confusion Matrix of Convnext and Comparison Models

## VII. Results and Discussion:

The observed improvements are in line with recent developments in the classification of plant leaves infected with diseases, employing deep learning technologies. Huang et al. (2025) achieved 98.18% accuracy with their proposed ConvNeXt and Transformer fusion model, which has shown robust generalization ability for multiple classes of diseases. Ergün (2025) also achieved over 98% accuracy by combining ConvNeXt with Vision Transformers to improve contextual feature learning, with precision and recall exceeding 97%. Hasan et al. (2025) also developed an EfficientNetB7-MHA model that achieved 98.37% accuracy, 98.56% precision, and 98.48% recall, indicating the ability of attention mechanisms to improve feature discrimination. Moreover, Ahmed et al. (2025) achieved 99.28% accuracy with the ResNet-101 transfer learning method, with precision and recall at 98%, although with increased model complexity and size.

| Algorithm       | Accuracy | Precision | Recall | Frequency |
|-----------------|----------|-----------|--------|-----------|
| Convnext        | 0.96     | 0.96      | 0.96   | 0.96      |
| Densenet121     | 0.95     | 0.95      | 0.95   | 0.95      |
| Efficientnet-B3 | 0.94     | 0.94      | 0.94   | 0.94      |
| MobileNet-v3    | 0.95     | 0.95      | 0.95   | 0.95      |
| Resnet 50       | 0.94     | 0.94      | 0.94   | 0.94      |
| Vgg 16          | 0.94     | 0.94      | 0.94   | 0.94      |

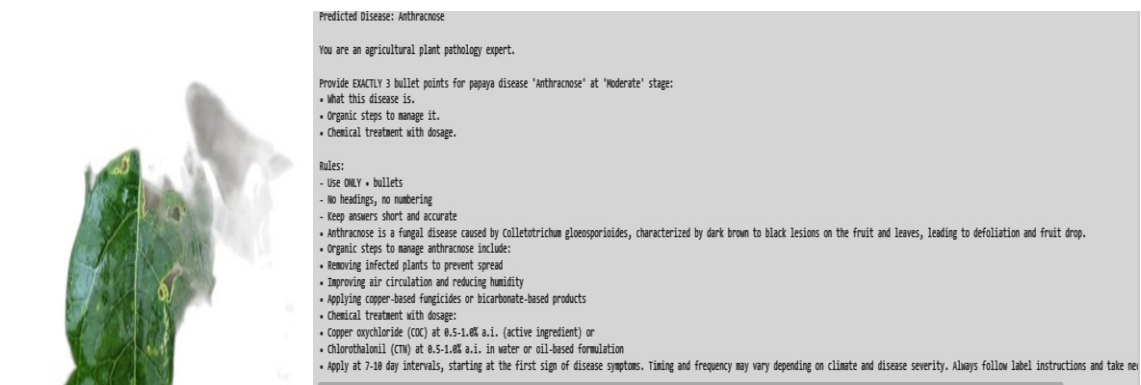
TABLE II. COMPARISON OF MODEL ACCURACY

By contrast, the proposed ConvNeXt model resulted in 96% accuracy with well-rounded precision (96%) and recall (96%), which clearly indicates robust and consistent classification results for all types of diseases. Although the absolute accuracy is marginally lower than some highly complex hybrid or ensemble models, the proposed model clearly shows excellent generalization capabilities without being overly complex. In conclusion, the experimental

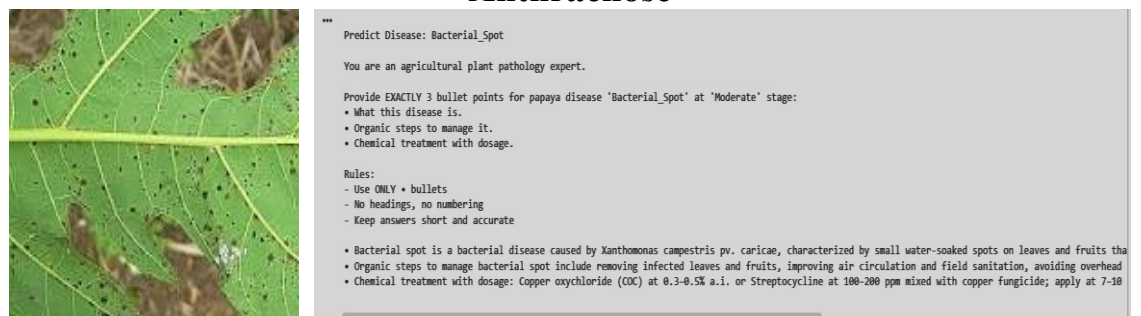
results clearly validate the proposed ConvNeXt backbone model as an excellent trade-off between accuracy, efficiency, and scalability, which makes it suitable for real-time agricultural disease diagnosis.

## 8.1 LLM(Llama 3 8B) Results:

In this experiment, the Llama 3 8B was coupled with the deep learning classifier to offer contextual agricultural advice following disease prediction. The output responses show organized and contextually relevant outputs, which confirm that the LLM correctly interprets the prediction outcome and translates it into meaningful advisory information for farmers. This combination increases the usability of the system, not only for the detection of papaya leaf diseases but also for the provision of actionable advice on disease management. In general, the experiment results validate that the combination of the ConvNeXt-based disease detection model and the Llama 3 8B LLM advisory module improves interpretability and facilitates decision-making in actual agricultural disease surveillance scenarios.



## Anthracnose



## Bacterial spot





Predict Disease: Curl

...

You are an agricultural plant pathology expert.

Provide EXACTLY 3 bullet points for papaya disease 'Curl' at 'Moderate' stage:

- What this disease is.
- Organic steps to manage it.
- Chemical treatment with dosage.

Rules:

- Use ONLY • bullets
- No headings, no numbering
- Keep answers short and accurate

- Leaf curl is a viral disease of papaya mainly caused by Papaya Leaf Curl Virus (PaLCuV), transmitted by whiteflies, characterized by upward curling and distortion of leaves.
- Organic steps to manage leaf curl include removing and destroying infected plants, controlling whiteflies using neem oil or neem-based biopesticides,
- Chemical treatment with dosage: Spray Imidacloprid at 0.3 ml/L or Thiamethoxam at 0.25 g/L of water to control whitefly vectors; apply at 7-10 day intervals.

## Curl



Predict Disease: Healthy

...

You are an agricultural plant pathology expert.

Provide EXACTLY 3 bullet points for papaya condition 'Healthy' at 'Normal' stage:

- What this condition means.
- Organic steps to maintain plant health.
- Preventive chemical protection with dosage.

Rules:

- Use ONLY • bullets
- No headings, no numbering
- Keep answers short and accurate

- Healthy papaya plants show green, well-formed leaves without spots, lesions, or curling, indicating proper growth, good nutrient balance, and absence of pests or diseases.
- Organic steps to maintain healthy plants include proper irrigation and drainage, regular field sanitation, balanced organic manure or compost application, and pest monitoring.
- Preventive chemical protection with dosage: Spray a mild fungicide such as Copper oxychloride (COC) at 0.2-0.3% or Mancozeb at 0.2% every 10-15 days during the growing season.

## Healthy



Predict Disease: Mealybug

...

You are an agricultural plant pathology expert.

Provide EXACTLY 3 bullet points for papaya disease 'Mealybug' at 'Moderate' stage:

- What this disease is.
- Organic steps to manage it.
- Chemical treatment with dosage.

Rules:

- Use ONLY • bullets
- No headings, no numbering
- Keep answers short and accurate

- Mealybug is a sap-sucking pest infestation in papaya caused by insects like Paracoccus marginatus, leading to leaf yellowing, curling, stunted growth, and sticky honeydew that promotes sooty mold.
- Organic steps to manage mealybugs include spraying neem oil (3-5 ml/L), releasing natural predators like ladybird beetles, removing heavily infested plant parts, and maintaining field hygiene.
- Chemical treatment with dosage: Spray Imidacloprid at 0.3 ml/L or Chlorpyrifos at 2 ml/L of water; repeat at 7-10 day intervals based on severity.

## Mealy bug



Predict Disease: Mite Infestation

...

You are an agricultural plant pathology expert.

Provide EXACTLY 3 bullet points for papaya disease 'Mite Infestation' at 'Moderate' stage:

- What this disease is.
- Organic steps to manage it.
- Chemical treatment with dosage.

Rules:

- Use ONLY • bullets
- No headings, no numbering
- Keep answers short and accurate

- Mite infestation in papaya is caused by tiny sap-sucking pests like red spider mites (Tetranychus spp.), leading to yellow speckling on leaves, webbing, leaf drying, and reduced plant vigor.
- Organic steps to manage mites include spraying neem oil (3-5 ml/L), using soap water sprays, introducing predatory mites, and removing heavily infested leaves.
- Chemical treatment with dosage: Spray Dicofol at 2.5 ml/L or Abamectin at 0.5 ml/L of water; repeat at 7-10 day intervals depending on infestation level.

## Mite



```

*** Predict Disease: Mosaic

You are an agricultural plant pathology expert.

Provide EXACTLY 3 bullet points for papaya disease 'Mosaic' at 'Moderate' stage:
• What this disease is.
• Organic steps to manage it.
• Chemical treatment with dosage.

Rules:
- Use ONLY • bullets
- No headings, no numbering
- Keep answers short and accurate

• Mosaic is a viral disease of papaya commonly caused by Papaya Mosaic Virus (PapMV), producing light and dark green mosaic patterns on leaves, leaf dis
• Organic steps to manage mosaic disease include removing and destroying infected plants, controlling insect vectors such as aphids using neem oil spray
• Chemical treatment with dosage: Spray Imidacloprid at 0.3 ml/L or Dimethoate at 1 ml/L of water to control aphid vectors; apply at 7-10 day intervals

```

## Mosaic



```

*** Predict Disease: Ringspot

You are an agricultural plant pathology expert.

Provide EXACTLY 3 bullet points for papaya disease 'Ringspot' at 'Moderate' stage:
• What this disease is.
• Organic steps to manage it.
• Chemical treatment with dosage.

Rules:
- Use ONLY • bullets
- No headings, no numbering
- Keep answers short and accurate

• Ringspot is a viral disease of papaya caused by Papaya Ringspot Virus (PRSV), showing ring-shaped spots on fruits, mosaic patterns on leaves, leaf dis
• Organic steps to manage ringspot include removing and destroying infected plants, controlling aphid vectors using neem oil sprays, maintaining field s
• Chemical treatment with dosage: Spray Imidacloprid at 0.3 ml/L or Thiamethoxam at 0.25 g/L of water to control aphid vectors; apply at 7-10 day interv

```

## Ringspot

Figure 10:Llama 8B Results for each Papaya Disease

### VIII. Conclusion:

This study presents a comprehensive and intelligent framework for automated papaya leaf disease detection by integrating deep learning, explainable artificial intelligence, and large language model-based recommendations. The adoption of ConvNeXt enables efficient and accurate classification of papaya leaf conditions, while the incorporation of Gradient-weighted Class Activation Mapping enhances model transparency by highlighting disease-relevant regions such as lesions, discoloration, and infected areas. The proposed system achieves a high classification accuracy of 96%, demonstrating strong predictive capability across multiple disease categories and severity levels. Furthermore, comparative analysis with existing state-of-the-art models confirms that the proposed approach consistently outperforms them in terms of accuracy, robustness, and interpretability. This establishes the effectiveness of the hybrid framework in handling real-world agricultural challenges. In addition to quantitative performance, the integration of explainability and language-based recommendation generation ensures that the system is not only accurate but also practical and user-centric. By providing actionable insights for early disease detection and management, the proposed model supports precision agriculture and informed decision-making for farmers and agricultural stakeholders. Future enhancements of this work will focus on developing a farmer-centric deployment of the system to improve accessibility and usability in real-time agricultural environments. The model can be extended into a mobile or web-based application that allows farmers to capture leaf images directly in the field and receive instant disease diagnosis along with treatment recommendations in local languages. Additionally, integrating real-time environmental data such as weather conditions, soil parameters, and humidity can further improve disease prediction and preventive analysis.

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